EXTRATERRESTRIAL GROUPS 

BASED ON WORK WITH K. BLOT, M. SABLIK AND V. SALO

Sebastián Barbieri

Universidad de Chile

June 3, 2026

What is this talk about?

Extraterrestrial group

Finitely generated group where (asymptotically) two “big” subsets which are close can be made far by removing a “small” number of elements.

- It’s a property of the large scale geometry (QI invariant).
- It has consequences for the computability of its actions.

What is this talk about?

Extraterrestrial group

Finitely generated group where (asymptotically) two “big” subsets which are close can be made far by removing a “small” number of elements.

- It’s a property of the large scale geometry (QI invariant).
- It has consequences for the computability of its actions.

Why are they relevant?

Theorem [BBSS]: Every extraterrestrial group admits an effective subshift which is not sofic.

What is this talk about?

Extraterrestrial group

Finitely generated group where (asymptotically) two “big” subsets which are close can be made far by removing a “small” number of elements.

- It’s a property of the large scale geometry (QI invariant).
- It has consequences for the computability of its actions.

Why are they relevant?

Theorem [BBSS]: Every extraterrestrial group admits an effective subshift which is not sofic.

What groups are extraterrestrial?

- Amenable groups.
- Multi-ended groups.
- Cocompact Fuchsian groups.
- Generalized Baumslag-Solitar groups.
- Thompson’s F is extraterrestrial if and only if it’s amenable.

BACKGROUND

- G is a finitely generated group.
- A is a finite set.
- The prodiscrete topology on A^G.
- The left shift action $G \curvearrowright A^G$.

$$(g \cdot x)(h) = x(g^{-1}h) \text{ for all } x \in A^G, g, h \in G.$$

BACKGROUND

- G is a finitely generated group.
- A is a finite set.
- The prodiscrete topology on A^G.
- The left shift action $G \curvearrowright A^G$.

$$(g \cdot x)(h) = x(g^{-1}h) \text{ for all } x \in A^G, g, h \in G.$$

Subshift

A **subshift** is a subset $X \subset A^G$ which is closed and G-invariant.

Background

- G is a finitely generated group.
- A is a finite set.
- The prodiscrete topology on A^G.
- The left shift action $G \curvearrowright A^G$.

$$(g \cdot x)(h) = x(g^{-1}h) \text{ for all } x \in A^G, g, h \in G.$$

Subshift

A **subshift** is a subset $X \subset A^G$ which is closed and G-invariant.

Types of subshift

A subshift X is called:

- **subshift of finite type** (SFT) if there is $F \Subset G$ such that $x \in X$ if and only if $gx \in X|_F$ for all $g \in G$.
- **sofic** if it is a topological factor of an SFT.
- **effective** if its complement $A^G \setminus X$ can be written as a union of cylinders enumerated by a Turing machine. ⚠️

Background

- G is a finitely generated group.
- A is a finite set.
- The prodiscrete topology on A^G.
- The left shift action $G \curvearrowright A^G$.

$$(g \cdot x)(h) = x(g^{-1}h) \text{ for all } x \in A^G, g, h \in G.$$

Subshift

A **subshift** is a subset $X \subset A^G$ which is closed and G-invariant.

Types of subshift

A subshift X is called:

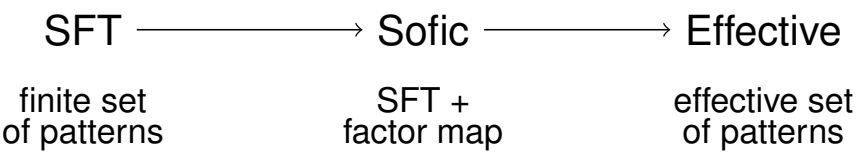
- **subshift of finite type** (SFT) if there is $F \Subset G$ such that $x \in X$ if and only if $gx \in X|_F$ for all $g \in G$.
- **sofic** if it is a topological factor of an SFT.
- **effective** if its complement $A^G \setminus X$ can be written as a union of cylinders enumerated by a Turing machine. ⚠️

⚠️ For simplicity, in this talk all groups have decidable word problem ⚠️

THE PROBLEM


SFT
Sofic
Effective
finite set
of patterns
SFT +
factor map
effective set
of patterns

THE PROBLEM

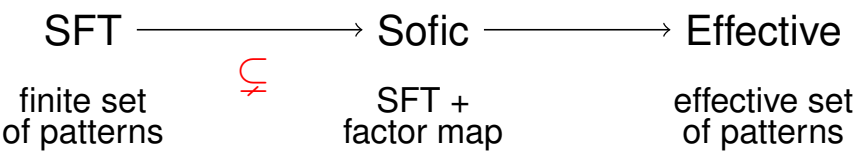


Folklore

In every infinite group, there are sofic subshifts which are not SFTs

- Even shift (with respect to a finite set which generates an infinite group).

THE PROBLEM

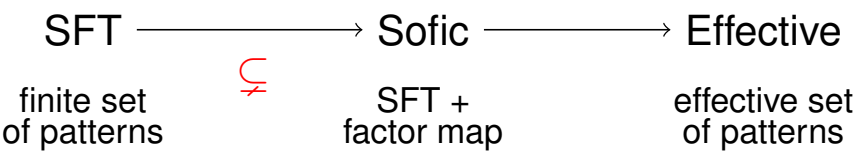


Folklore

In every infinite group, there are sofic subshifts which are not SFTs

- Even shift (with respect to a finite set which generates an infinite group).

THE PROBLEM



Folklore

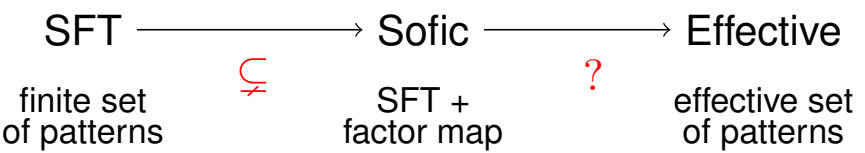
In every infinite group, there are sofic subshifts which are not SFTs

- Even shift (with respect to a finite set which generates an infinite group).

Question

For which infinite groups are there effective but non-sofic subshifts?

THE PROBLEM



Folklore

In every infinite group, there are sofic subshifts which are not SFTs

- Even shift (with respect to a finite set which generates an infinite group).

Question

For which infinite groups are there effective but non-sofic subshifts?

Effective vs sofic

Groups with effective subshifts which are not sofic

- $\mathbb{Z}$: Sturmian with algebraic non-rational slope.

Effective vs sofic

Groups with effective subshifts which are not sofic

- $\mathbb{Z}$: Sturmian with algebraic non-rational slope.
- $\mathbb{Z}^2$: Mirror shift.

Effective vs sofic

Groups with effective subshifts which are not sofic

- $\mathbb{Z}$: Sturmian with algebraic non-rational slope.
- $\mathbb{Z}^2$: Mirror shift.
- Amenable groups: variation of the mirror shift.

Effective vs sofic

Groups with effective subshifts which are not sofic

- $\mathbb{Z}$: Sturmian with algebraic non-rational slope.
- $\mathbb{Z}^2$: Mirror shift.
- Amenable groups: variation of the mirror shift.
- Multi-ended groups: variation of the mirror shift.

Effective vs sofic

Groups with effective subshifts which are not sofic

- $\mathbb{Z}$: Sturmian with algebraic non-rational slope.
- $\mathbb{Z}^2$: Mirror shift.
- Amenable groups: variation of the mirror shift.
- Multi-ended groups: variation of the mirror shift.

Back in 2014, I thought there were effective but non-sofic shifts on every infinite group.

Effective vs sofic

Groups with effective subshifts which are not sofic

- $\mathbb{Z}$: Sturmian with algebraic non-rational slope.
- $\mathbb{Z}^2$: Mirror shift.
- Amenable groups: variation of the mirror shift.
- Multi-ended groups: variation of the mirror shift.

Back in 2014, I thought there were effective but non-sofic shifts on every infinite group.

I was WRONG 😵.

Effective vs sofic

Groups with effective subshifts which are not sofic

- $\mathbb{Z}$: Sturmian with algebraic non-rational slope.
- $\mathbb{Z}^2$: Mirror shift.
- Amenable groups: variation of the mirror shift.
- Multi-ended groups: variation of the mirror shift.

Back in 2014, I thought there were effective but non-sofic shifts on every infinite group.

I was WRONG 😵.

Self-simulable group

A group is **self-simulable** if every effective subshift is sofic.

EFFECTIVE VS SOFIC

Groups with effective subshifts which are not sofic

- $\mathbb{Z}$: Sturmian with algebraic non-rational slope.
- $\mathbb{Z}^2$: Mirror shift.
- Amenable groups: variation of the mirror shift.
- Multi-ended groups: variation of the mirror shift.

Back in 2014, I thought there were effective but non-sofic shifts on every infinite group.

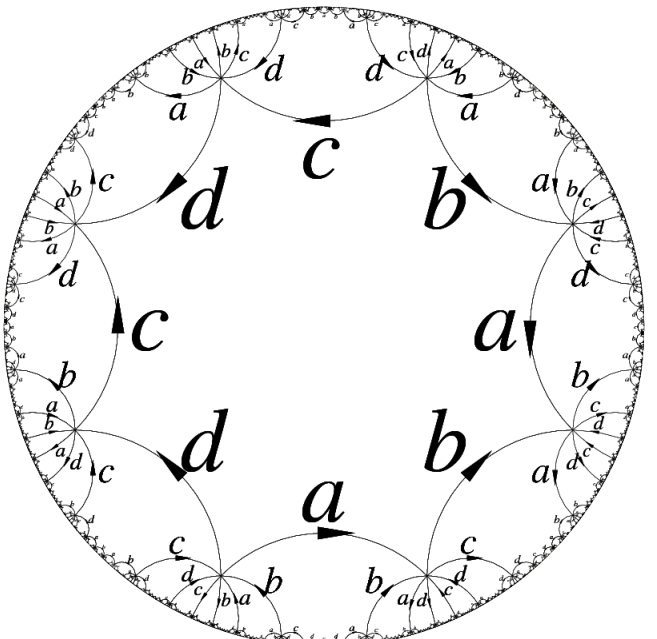
I was WRONG 😵.

Self-simulable group

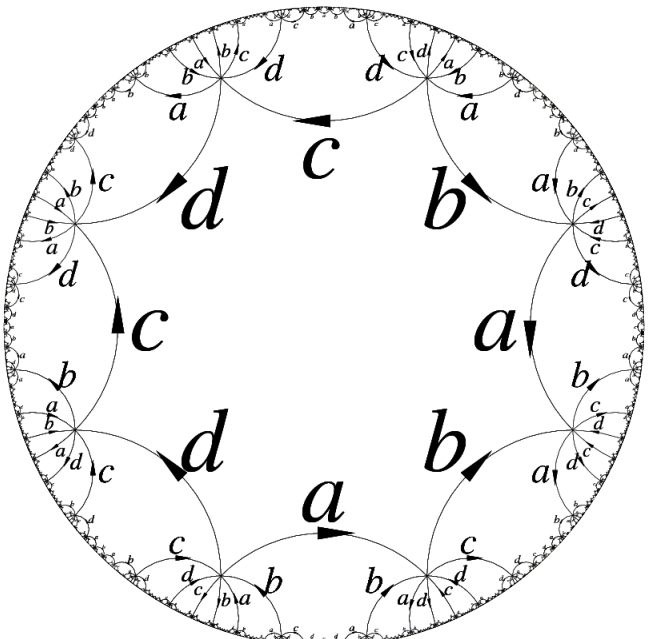
A group is **self-simulable** if every effective subshift is sofic.

- $F_2 \times F_2$ (the product of any two nonamenable groups).
- $\mathrm{SL}_n(\mathbb{Z})$, $n \geq 5$.
- $\mathrm{Aut}(F_n)$, $n \geq 5$.
- Braid groups B_n, $n \geq 7$.
- Thompson's V.

Are surface groups self-simulable?



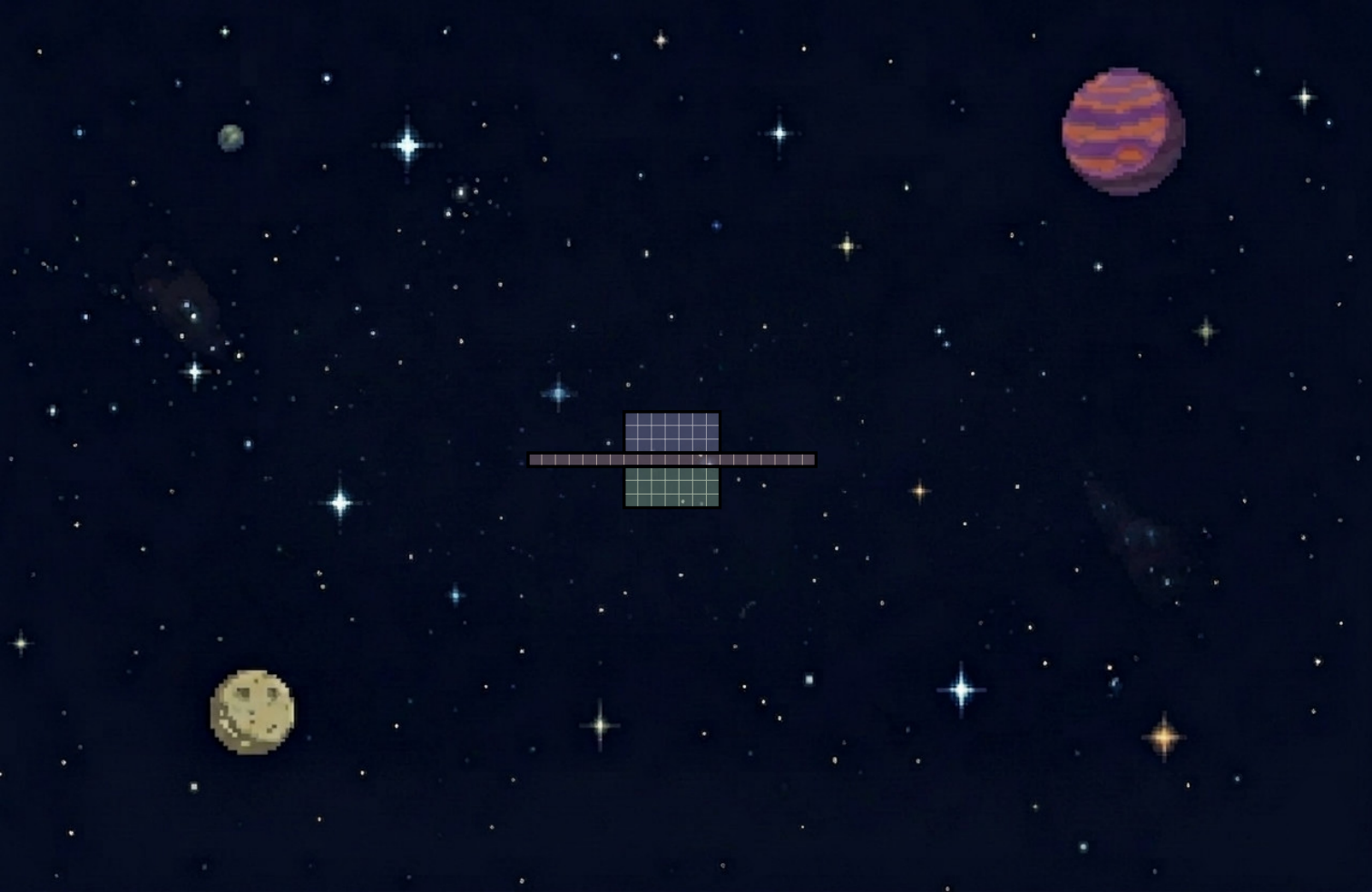
ARE SURFACE GROUPS SELF-SIMULABLE?

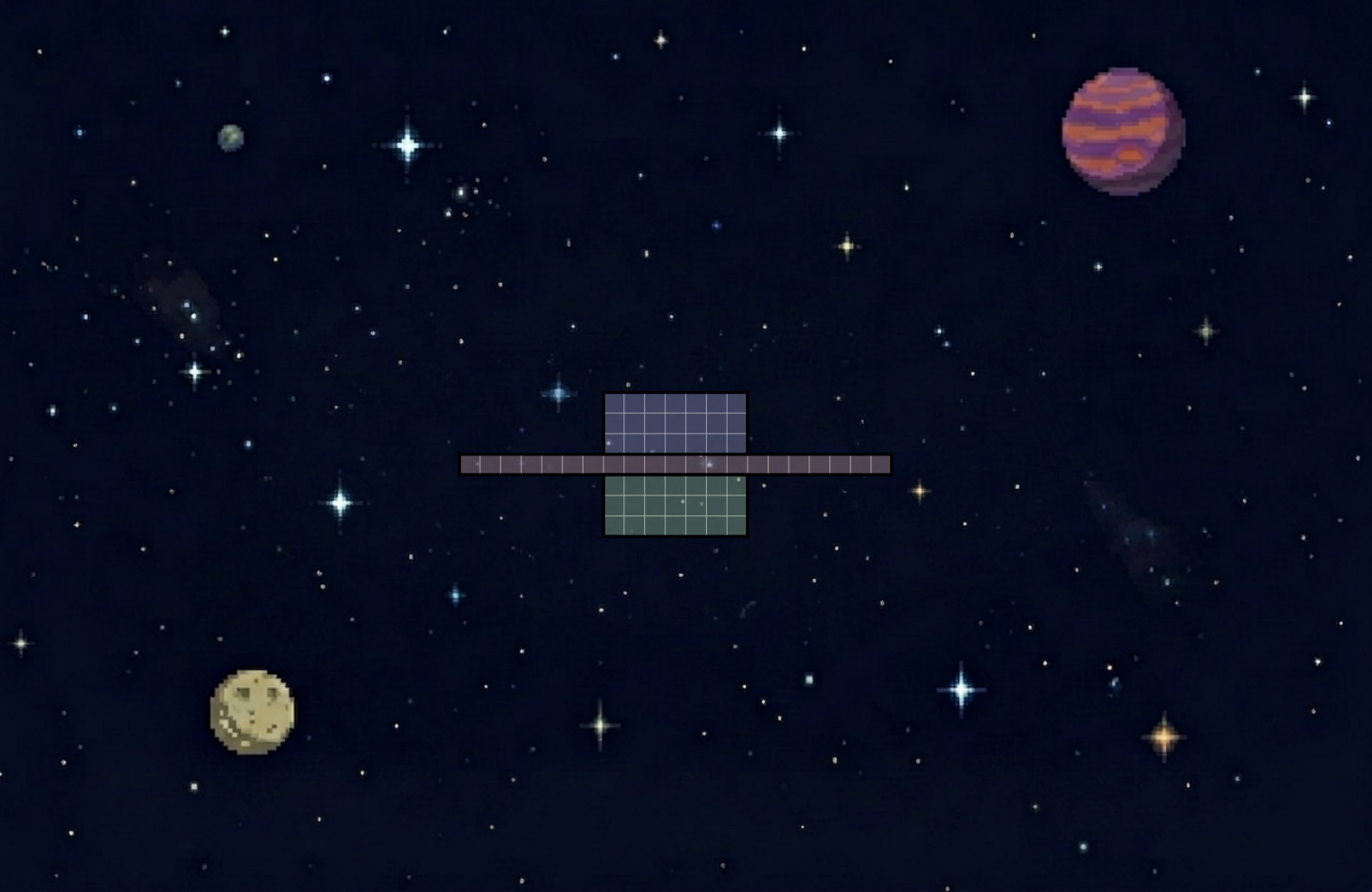


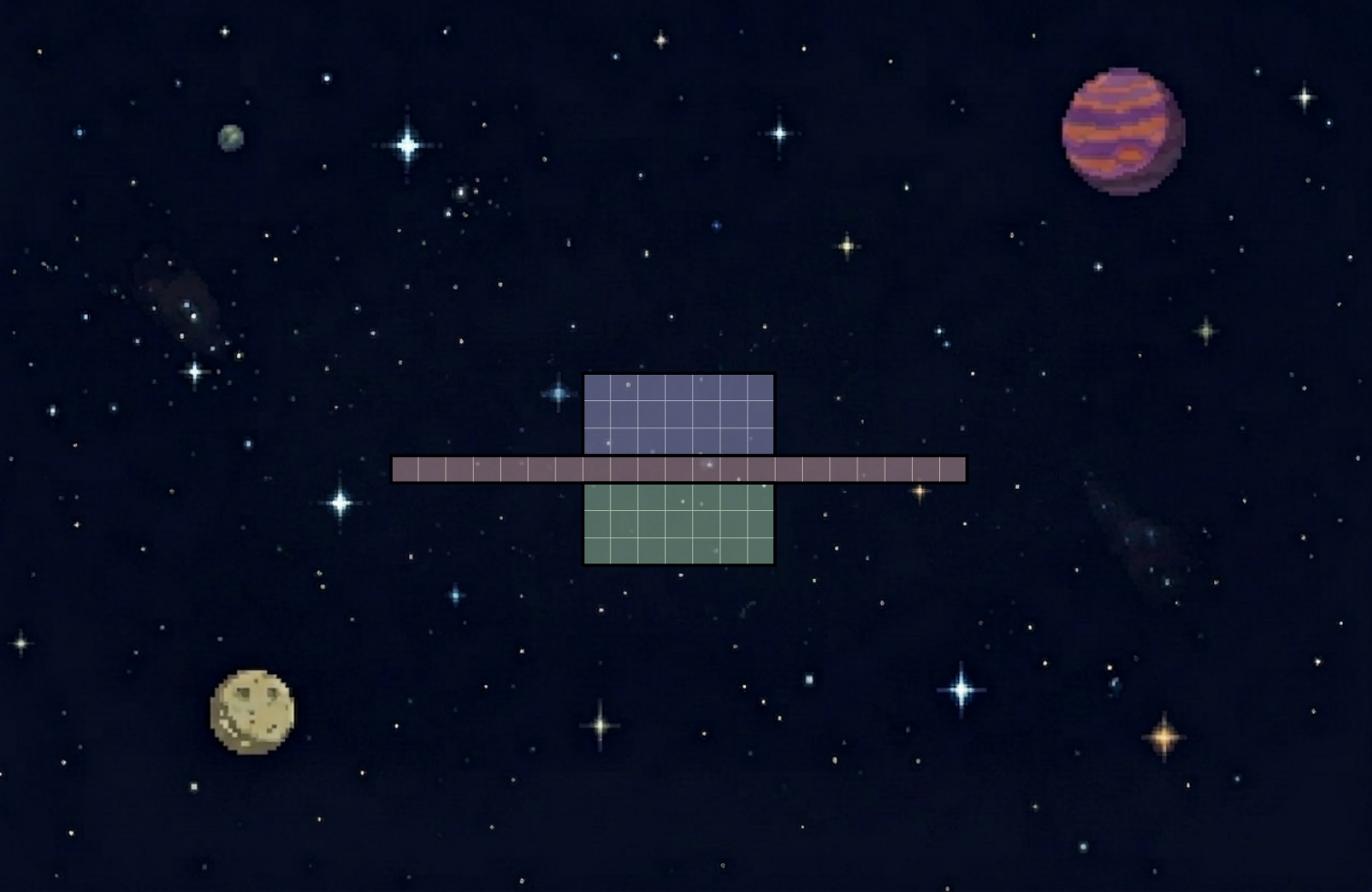
“The answer you seek resides in the stars above”
Kung Pow! enter the fist, 2002.

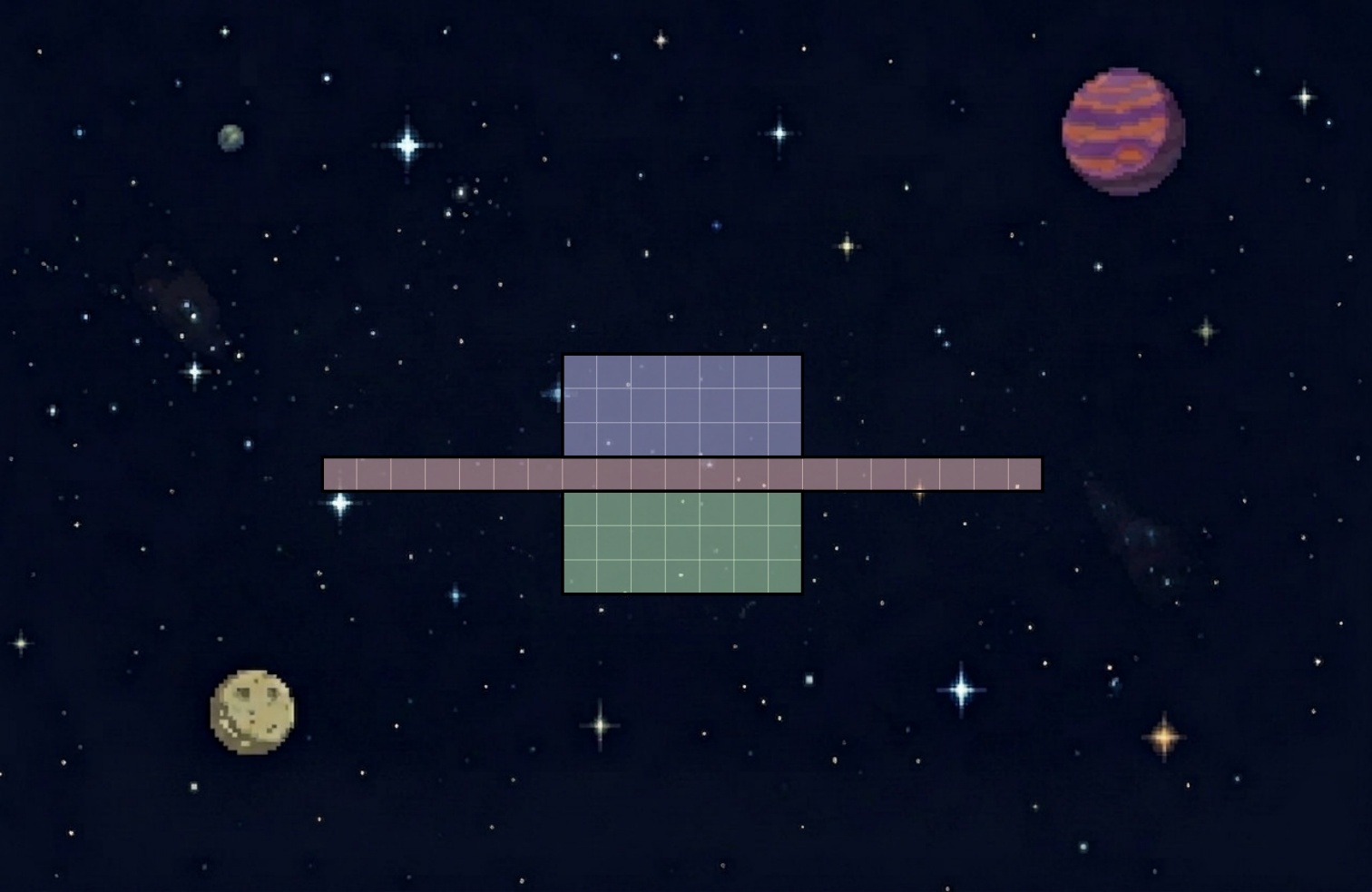


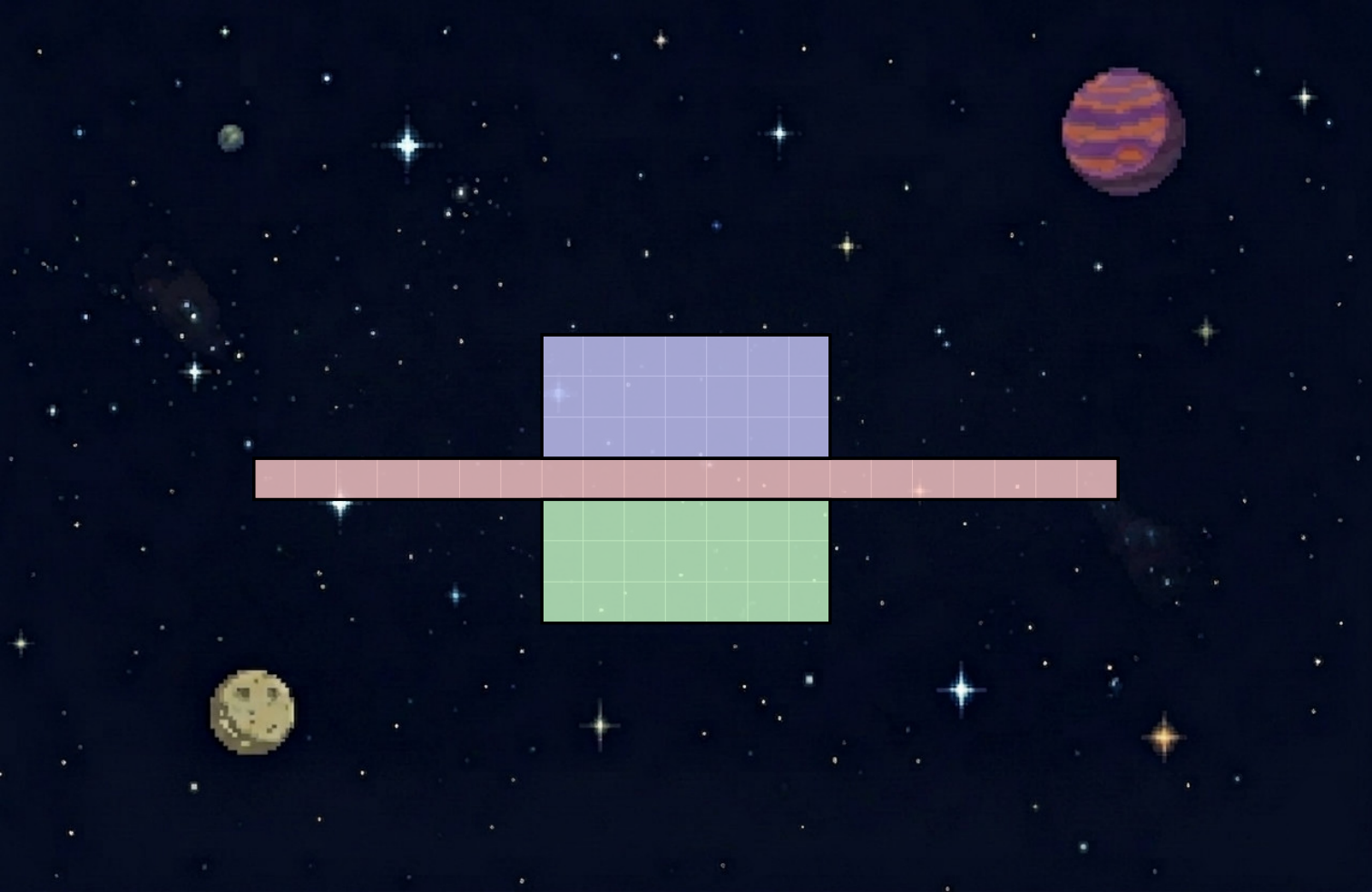












EXTRATERRESTRIAL GROUPS

- $\Gamma = (V, E)$ is a countable, locally finite graph.
- m, k, r are positive integers.

EXTRATERRESTRIAL GROUPS

- $\Gamma = (V, E)$ is a countable, locally finite graph.
- m, k, r are positive integers.

UFO

An (m, k, r)-**UFO** in Γ is a triple (U, F, O) of nonempty pairwise disjoint finite sets of vertices in V such that:

1. $|U| = |O| = m|F|$.
2. There is a perfect matching between U and O with paths of length at most k.
3. Any path from U to O that avoids vertices in F has length at least r.

EXTRATERRESTRIAL GROUPS

- $\Gamma = (V, E)$ is a countable, locally finite graph.
- m, k, r are positive integers.

UFO

An (m, k, r)-**UFO** in Γ is a triple (U, F, O) of nonempty pairwise disjoint finite sets of vertices in V such that:

1. $|U| = |O| = m|F|$.
2. There is a perfect matching between U and O with paths of length at most k.
3. Any path from U to O that avoids vertices in F has length at least r.

Extraterrestrial graph

A graph Γ is called **extraterrestrial** if for every m, there is k such that for every r there is some (m, k, r)-UFO.

EXTRATERRESTRIAL GROUPS

- $\Gamma = (V, E)$ is a countable, locally finite graph.
- m, k, r are positive integers.

UFO

An (m, k, r)-**UFO** in Γ is a triple (U, F, O) of nonempty pairwise disjoint finite sets of vertices in V such that:

1. $|U| = |O| = m|F|$.
2. There is a perfect matching between U and O with paths of length at most k.
3. Any path from U to O that avoids vertices in F has length at least r.

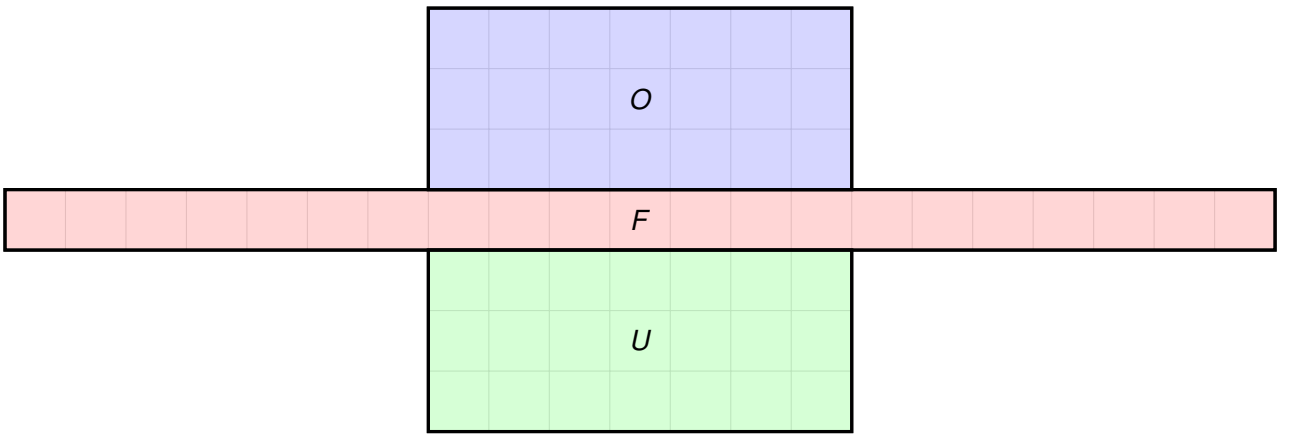
Extraterrestrial graph

A graph Γ is called **extraterrestrial** if for every m, there is k such that for every r there is some (m, k, r)-UFO.

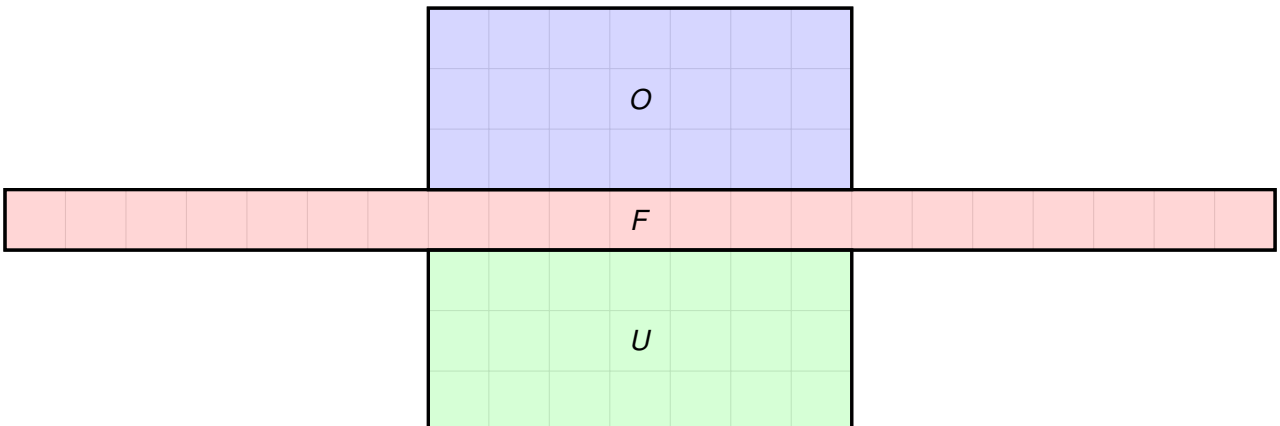
Extraterrestrial group

A finitely generated group G is called **extraterrestrial** if $\mathrm{Cay}(G, S)$ is extraterrestrial for some set of generators S.

Extraterrestrial groups

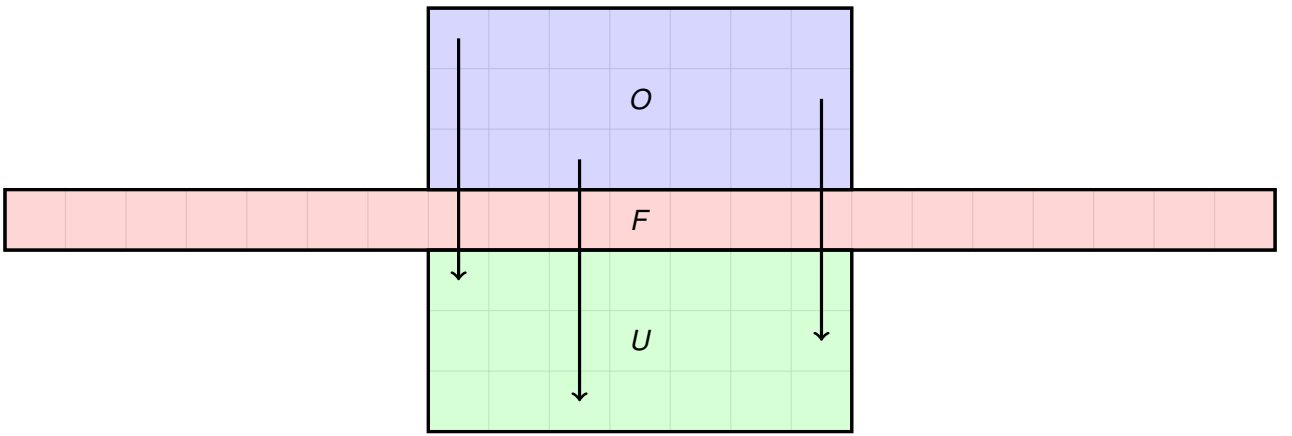


EXTRATERRESTRIAL GROUPS



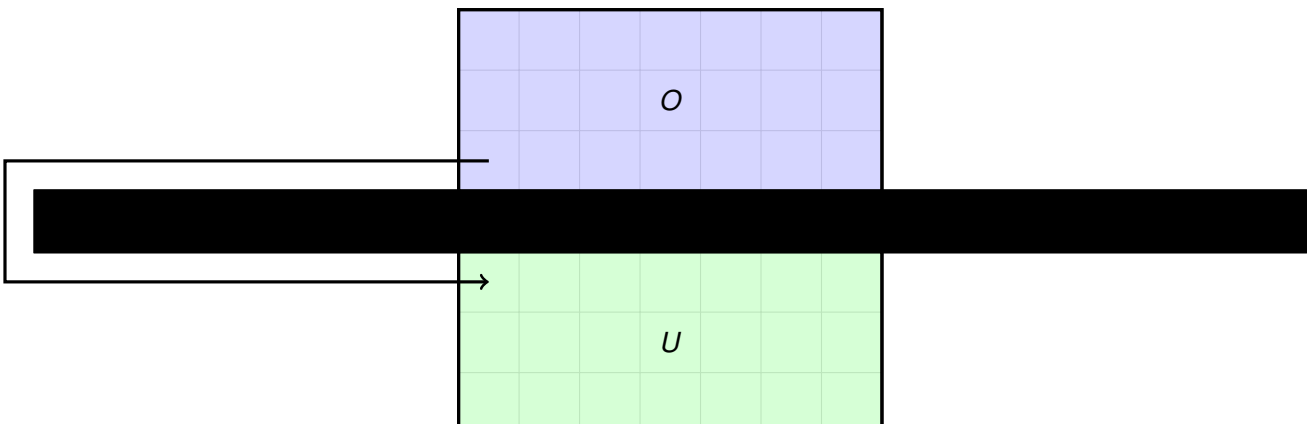
1. $|O| = |U| = 1|F| \longrightarrow m = 1.$

EXTRATERRESTRIAL GROUPS



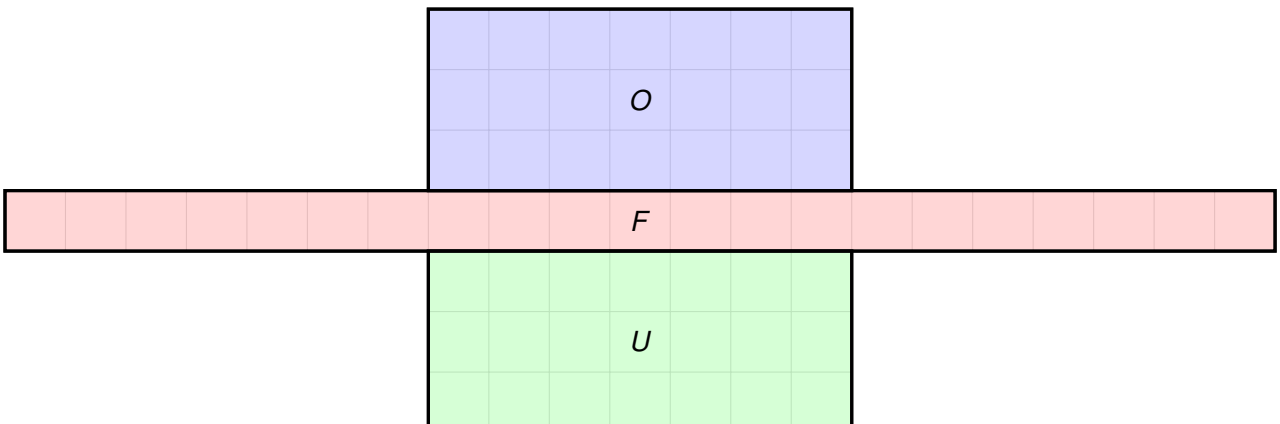
1. $|O| = |U| = 1|F| \longrightarrow m = 1$.
2. There is a matching from U to O with paths of length $4 \longrightarrow k = 4$.

EXTRATERRESTRIAL GROUPS



1. $|O| = |U| = 1|F| \longrightarrow m = 1$.
2. There is a matching from U to O with paths of length $4 \longrightarrow k = 4$.
3. All paths from U to O avoiding F have length at least $18 \longrightarrow r = 18$.

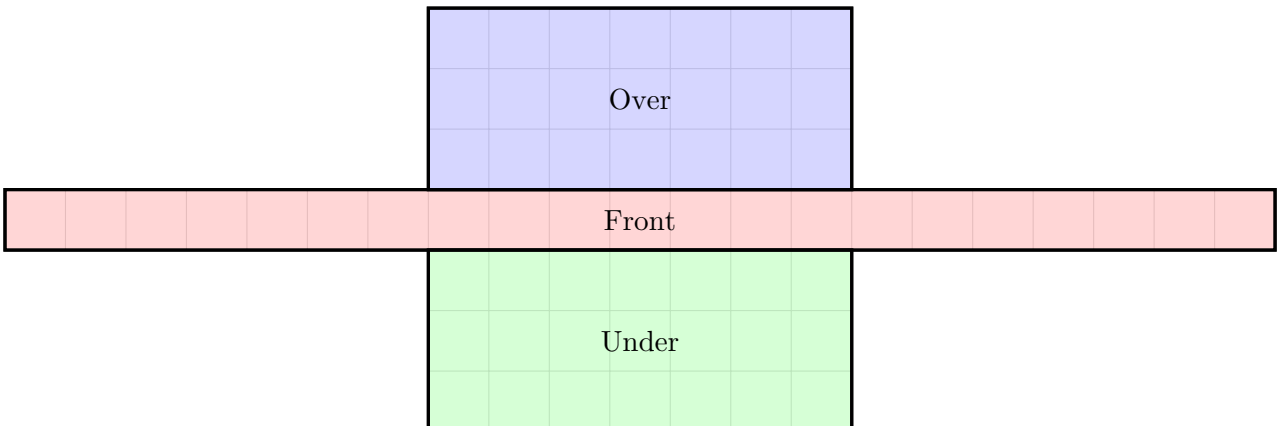
EXTRATERRESTRIAL GROUPS



1. $|O| = |U| = 1|F| \longrightarrow m = 1$.
2. There is a matching from U to O with paths of length $4 \longrightarrow k = 4$.
3. All paths from U to O avoiding F have length at least $18 \longrightarrow r = 18$.

(U, F, O) is a $(1, 4, 18)$-UFO in the canonical Cayley graph of $\mathbb{Z}^2$.

Extraterrestrial groups



1. $|O| = |U| = 1|F| \longrightarrow m = 1$.
2. There is a matching from U to O with paths of length $4 \longrightarrow k = 4$.
3. All paths from U to O avoiding F have length at least $18 \longrightarrow r = 18$.

(U, F, O) is a $(1, 4, 18)$-UFO in the canonical Cayley graph of $\mathbb{Z}^2$.

Quasi-isometry

Given metric spaces (X, d_X) and (Y, d_Y), a map $f\colon X \to Y$ is a **quasi-isometry** if there is $C \geq 1$ such that

- (QI embedding) For all $x, x' \in X$, $\frac{1}{C} d_X(x, x') - C \leq d_Y(f(x), f(x')) \leq C d_X(x, x') + C$.
- (relatively dense) For every $y \in Y$ there is $x \in X$ such that $d_Y(f(x), y) \leq C$.

Quasi-isometry

Given metric spaces (X, d_X) and (Y, d_Y), a map $f\colon X \to Y$ is a **quasi-isometry** if there is $C \geq 1$ such that

- (QI embedding) For all $x, x' \in X$, $\frac{1}{C} d_X(x, x') - C \leq d_Y(f(x), f(x')) \leq C d_X(x, x') + C$.
- (relatively dense) For every $y \in Y$ there is $x \in X$ such that $d_Y(f(x), y) \leq C$.

Theorem

Being extraterrestrial is a quasi-isometry invariant for graphs of bounded degree.

Quasi-isometry

Given metric spaces (X, d_X) and (Y, d_Y), a map $f\colon X \to Y$ is a **quasi-isometry** if there is $C \geq 1$ such that

- (QI embedding) For all $x, x' \in X$, $\frac{1}{C} d_X(x, x') - C \leq d_Y(f(x), f(x')) \leq C d_X(x, x') + C$.
- (relatively dense) For every $y \in Y$ there is $x \in X$ such that $d_Y(f(x), y) \leq C$.

Theorem

Being extraterrestrial is a quasi-isometry invariant for graphs of bounded degree.

Observations

1. Being extraterrestrial is an honest property of a group. It does not depend on the set of generators.

Quasi-isometry

Given metric spaces (X, d_X) and (Y, d_Y), a map $f\colon X \to Y$ is a **quasi-isometry** if there is $C \geq 1$ such that

- (QI embedding) For all $x, x' \in X$, $\frac{1}{C} d_X(x, x') - C \leq d_Y(f(x), f(x')) \leq C d_X(x, x') + C$.
- (relatively dense) For every $y \in Y$ there is $x \in X$ such that $d_Y(f(x), y) \leq C$.

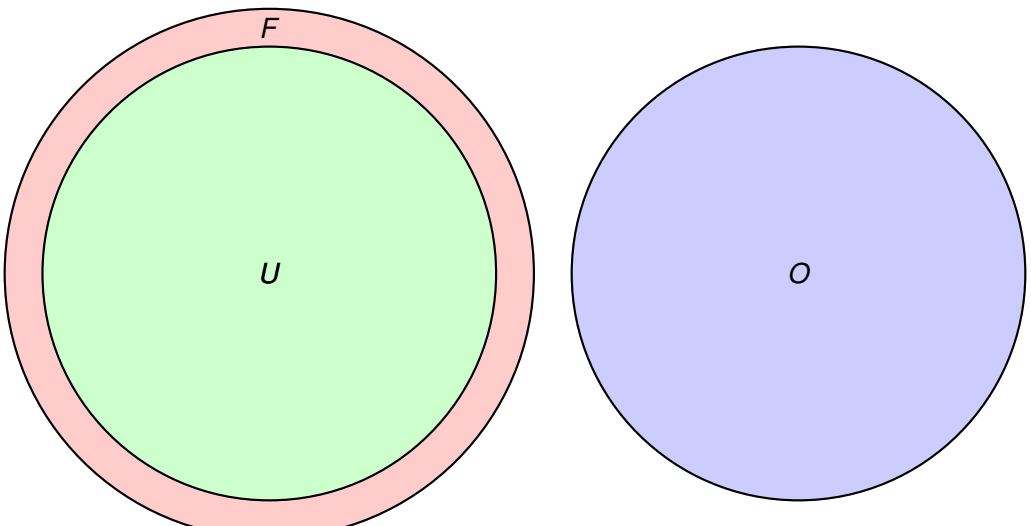
Theorem

Being extraterrestrial is a quasi-isometry invariant for graphs of bounded degree.

Observations

1. Being extraterrestrial is an honest property of a group. It does not depend on the set of generators.
2. To prove a group is extraterrestrial, it suffices to do it for some graph which is quasi-isometric to it.

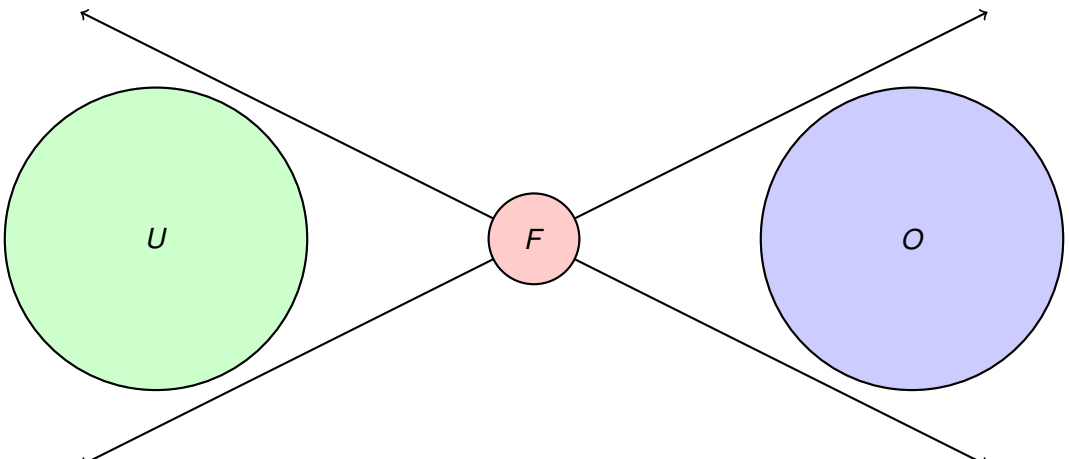
AMENABLE GROUPS ARE EXTRATERRESTRIAL



Choose $m \geq 1$. Take U and $F = \partial U$ such that $|U| \geq m|\partial U|$ and choose O as any set with $|U|$ elements disjoint from $U \cup F$.

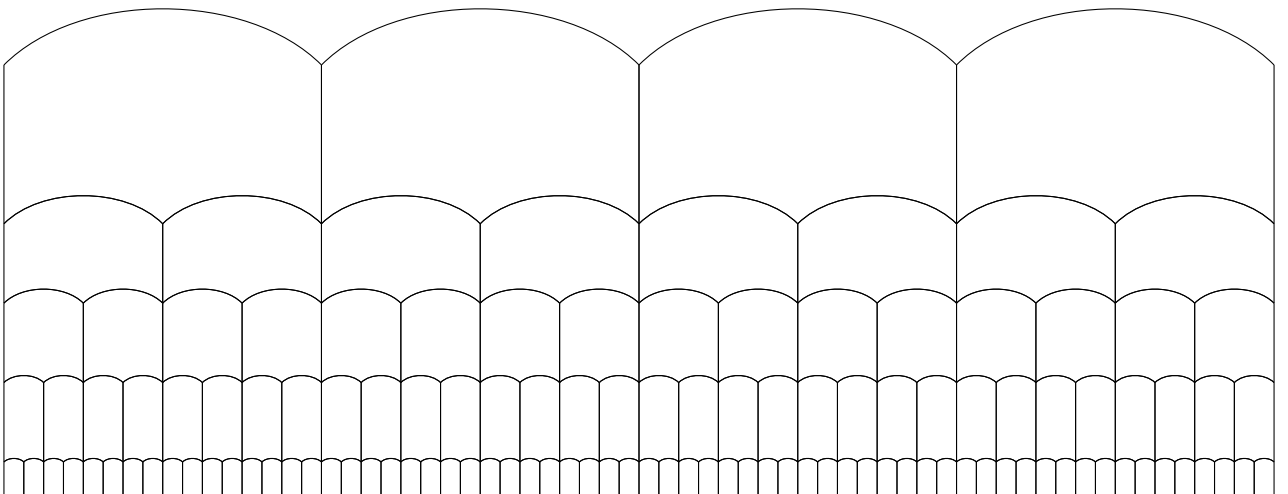
Take $k = \max d(g, h) : g \in U, h \in O$, then (U, F, O) is an (m, k, r)-UFO for all $r \geq 0$.

MULTI-ENDED GROUPS ARE EXTRATERRESTRIAL

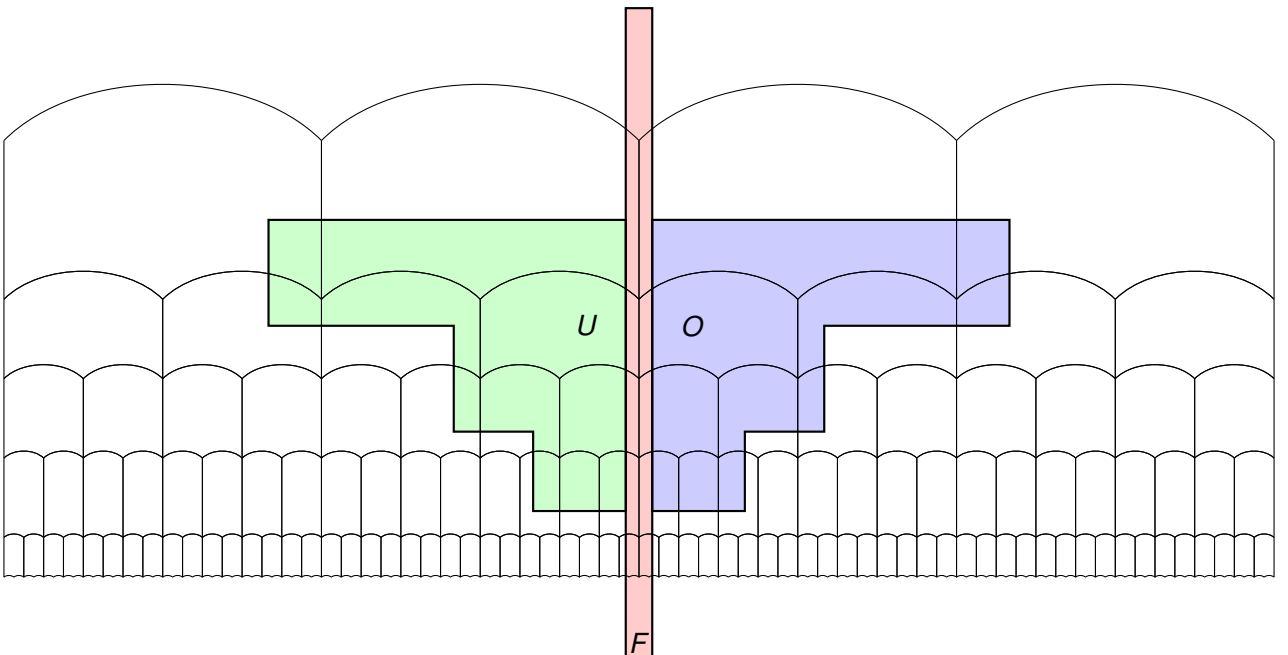


Choose F so that it cuts G in at least two infinite connected components. Choose U and O in different components as large as you want.

Cocompact Fuchsian groups are extraterrestrial



COCOMPACT FUCHSIAN GROUPS ARE EXTRATERRESTRIAL



EXAMPLES OF EXTRATERRESTRIAL GROUPS

The following groups are extraterrestrial

1. Amenable groups.
2. Multi-ended groups.
3. Cocompact Fuchsian groups.
4. Fundamental groups of surfaces.
5. If K is amenable, H is extraterrestrial and $1 \to K \to G \to H \to 1$, then G is extraterrestrial.
6. $F_2 \times \mathbb{Z}$ is extraterrestrial.
7. Amalgamated free products relative to an amenable subgroup with strict embeddings.
8. HNN-extensions relative to an amenable subgroup.
9. Generalized Baumslag Solitar groups.

Extraterrestrial groups are not self-simulable

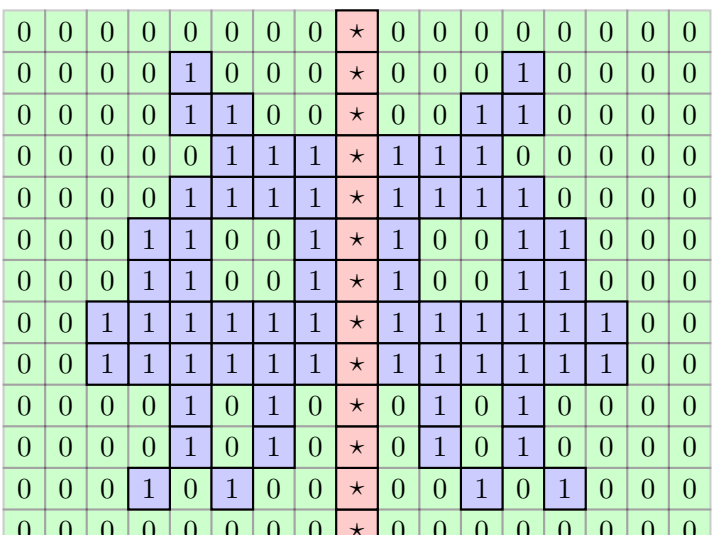
Main idea

Generalize the mirror shift in a crazy way.

EXTRATERRESTRIAL GROUPS ARE NOT SELF-SIMULABLE

Main idea

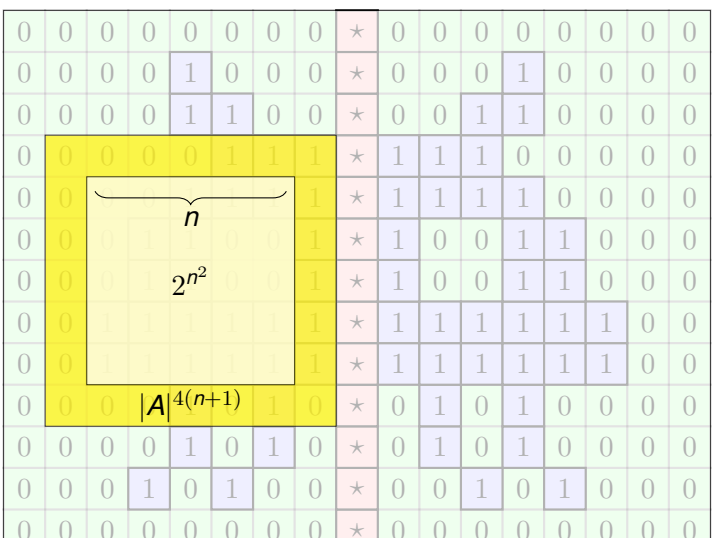
Generalize the mirror shift in a crazy way.



Extraterrestrial groups are not self-simulable

Main idea

Generalize the mirror shift in a crazy way.



EXTRATERRESTRIAL GROUPS ARE NOT SELF-SIMULABLE

Ingredients

1. ⚠️ G has decidable word problem.
2. Alphabet $A = \{\mathfrak{u}, \mathfrak{f}, \mathfrak{o}\} \times \{0, 1\}$.
3. Fix a computable total well order on G.

EXTRATERRESTRIAL GROUPS ARE NOT SELF-SIMULABLE

Ingredients

1. ⚠️ G has decidable word problem.
2. Alphabet $A = \{\mathfrak{u}, \mathfrak{f}, \mathfrak{o}\} \times \{0, 1\}$.
3. Fix a computable total well order on G.

For a configuration $\xi \in \{\mathfrak{u}, \mathfrak{o}, \mathfrak{f}\}^G$, match greedily positions with symbols $\mathfrak{u}$ to positions with symbols $\mathfrak{o}$. This defines a unique partial matching $M(\xi)$.

EXTRATERRESTRIAL GROUPS ARE NOT SELF-SIMULABLE

Ingredients

1. ⚠️ G has decidable word problem.
2. Alphabet $A = \{\mathfrak{u}, \mathfrak{f}, \mathfrak{o}\} \times \{0, 1\}$.
3. Fix a computable total well order on G.

For a configuration $\xi \in \{\mathfrak{u}, \mathfrak{o}, \mathfrak{f}\}^G$, match greedily positions with symbols $\mathfrak{u}$ to positions with symbols $\mathfrak{o}$. This defines a unique partial matching $M(\xi)$.

The super mega generalized mirror shift

All configurations in $(\xi, v) \in A^G$ such that if g is matched with h in $M(\xi)$, then $v(g) = v(h)$.

EXTRATERRESTRIAL GROUPS ARE NOT SELF-SIMULABLE

Ingredients

1. ⚠️ G has decidable word problem.
2. Alphabet $A = \{\mathfrak{u}, \mathfrak{f}, \mathfrak{o}\} \times \{0, 1\}$.
3. Fix a computable total well order on G.

For a configuration $\xi \in \{\mathfrak{u}, \mathfrak{o}, \mathfrak{f}\}^G$, match greedily positions with symbols $\mathfrak{u}$ to positions with symbols $\mathfrak{o}$. This defines a unique partial matching $M(\xi)$.

The super mega generalized mirror shift

All configurations in $(\xi, v) \in A^G$ such that if g is matched with h in $M(\xi)$, then $v(g) = v(h)$.

Observation

The super mega generalized mirror shift is effective.

EXTRATERRESTRIAL GROUPS ARE NOT SELF-SIMULABLE

Theorem

If G is extraterrestrial, then the super mega generalized mirror shift is not sofic.

EXTRATERRESTRIAL GROUPS ARE NOT SELF-SIMULABLE

Theorem

If G is extraterrestrial, then the super mega generalized mirror shift is not sofic.

Proof sketch

1. Suppose there is a nn SFT extension X to the SMGMS Y through a 1-block map.
2. Let q be the size of the alphabet of X.
3. Fix $m > 2\log(q)$. There is k, such that for all r there is an (m, k, r) UFO (U, F, O) on G.
4. Declare ξ_r as $\mathfrak{u}$ on U, $\mathfrak{o}$ in O and $\mathfrak{f}$ everywhere else.
5. Counting argument: there are $x_r^1, x_r^2 \in X$, which map to (ξ_r, y_r^1) and (ξ_r, y_r^2) such that $x_r^1|_F = x_r^2|_F$ but y_r^1, y_r^2 differ in some matched symbol at distance k.
6. Shift appropriately and take limit points as $r \to \infty$. You get $x^1, x^2 \in X$ which coincide in an infinite set F which cuts U from O, and such that their $0/1$ values differ in some matched pair of elements at distance k near the identity.
7. Cut and paste in the SFT, project using the 1-block map, get a point outside of Y.

Final remarks

Observation

The proof without the assumption of decidable word problem becomes way more complicated. The subshift above is not effective.

Some ideas:

1. We need to consider relative effective subshifts by taking pullbacks in the free group and “trick” the group into simulating an uncomputable word problem.
2. We use a technical lemma which might be of independent interest.

FINAL REMARKS

Observation

The proof without the assumption of decidable word problem becomes way more complicated. The subshift above is not effective.

Some ideas:

1. We need to consider relative effective subshifts by taking pullbacks in the free group and “trick” the group into simulating an uncomputable word problem.
2. We use a technical lemma which might be of independent interest.

Lemma

There exists a universal constant $A \geq 1$ such that for every infinite and finitely generated group G, and every finite generating set S, there exists a configuration $\xi \in \{0,1\}^G$ such that for every integer $k \geq 1$ and $g, h \in G$ with $d_S(g,h) = k$, there is $t \in G$ with $|t|_S \leq kA$ such that $\xi(gt) \neq \xi(ht)$.

FINAL REMARKS

Observation

The proof without the assumption of decidable word problem becomes way more complicated. The subshift above is not effective.

Some ideas:

1. We need to consider relative effective subshifts by taking pullbacks in the free group and “trick” the group into simulating an uncomputable word problem.
2. We use a technical lemma which might be of independent interest.

Lemma

There exists a universal constant $A \geq 1$ such that for every infinite and finitely generated group G, and every finite generating set S, there exists a configuration $\xi \in \{0,1\}^G$ such that for every integer $k \geq 1$ and $g, h \in G$ with $d_S(g,h) = k$, there is $t \in G$ with $|t|_S \leq kA$ such that $\xi(gt) \neq \xi(ht)$.

Believe it or not, the best constant we know is $A = 51$ 🛸.

Final remarks

Every group which we know is not self-simulable, happens to be extraterrestrial.

Question 1

Is there a group which is neither extraterrestrial nor self-simulable?

FINAL REMARKS

Every group which we know is not self-simulable, happens to be extraterrestrial.

Question 1

Is there a group which is neither extraterrestrial nor self-simulable?

Thompson's group F is the group of all piecewise linear orientation-preserving homeomorphisms of $[0, 1]$ with dyadic breakpoints. It is a famous open problem whether F is amenable or not.

Remark

From previous work, we know that F is nonamenable if and only if it is self-simulable.

- If F were amenable, then it would be extraterrestrial.
- If F were extraterrestrial, it would not be self-simulable, and thus it would be amenable.
- Hence F is amenable if and only if it is extraterrestrial.

FINAL REMARKS

Every group which we know is not self-simulable, happens to be extraterrestrial.

Question 1

Is there a group which is neither extraterrestrial nor self-simulable?

Thompson's group F is the group of all piecewise linear orientation-preserving homeomorphisms of $[0, 1]$ with dyadic breakpoints. It is a famous open problem whether F is amenable or not.

Remark

From previous work, we know that F is nonamenable if and only if it is self-simulable.

- If F were amenable, then it would be extraterrestrial.
- If F were extraterrestrial, it would not be self-simulable, and thus it would be amenable.
- Hence F is amenable if and only if it is extraterrestrial.

Question 2

Is Thompson's F extraterrestrial?


Thank you for your attention